

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 1

**Duration :60 Mins
On Class
Total Marks:50**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Ranjitha. R (192421285) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:





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Language Processors: Compiler vs Interpreter

CSA1408 - Compiler Design

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BY:

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Language Processor



- A language processor is software that converts a high-level programming language into machine language.
- It helps the computer understand and execute programs.

Types of Language Processors

- Compiler
- Interpreter
- Assembler

Examples of High-Level Languages:

- C
- C++
- Java
- Python



Compiler



- A compiler translates the **entire program** into machine code before execution.

Features :

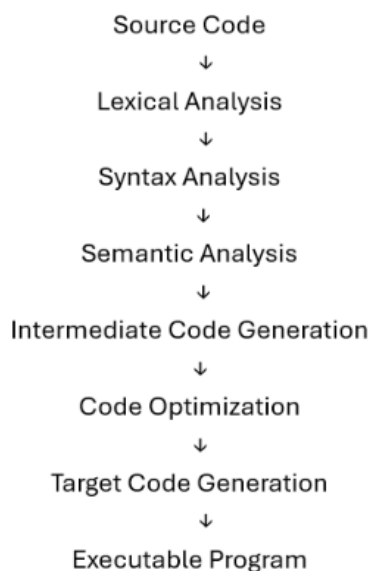
- Converts the whole program at once.
- Generates an executable file.
- Executes programs faster.
- Shows all errors after compilation.

Examples

- C Compiler (GCC)
- C++ Compiler (G++)



Compiler



compiler optimization techniques used to make programs run faster, use less memory, and improve efficiency without changing the program's output

Important Optimizations

- Constant Folding
- Dead Code Elimination
- Loop Optimization
- Function Inlining
- Register Allocation
- Common Subexpression Elimination

Interpreter

- An interpreter translates and executes the program **one line at a time**.

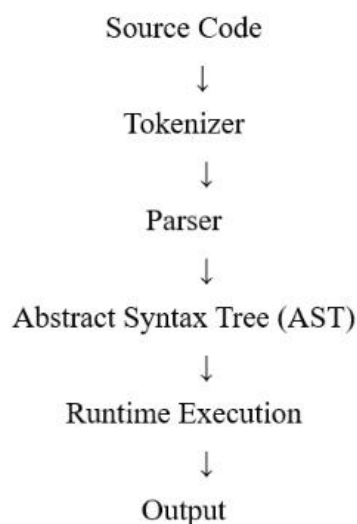
Features

- No executable file is created.
- Executes line by line.
- Stops when an error is found.
- Easier for debugging but slower than a compiler.

Examples

- Python Interpreter
- JavaScript Engine

Interpreter



Characteristics

- Immediate execution without generating an executable.
- Runtime type checking.
- Dynamic memory allocation.
- Interactive debugging support.
- High flexibility for scripting and rapid development.

Performance Challenges

- Repeated parsing.
- Higher CPU overhead.
- Slower execution for computationally intensive applications.



Compiler vs Interpreter



Compiler	Interpreter
Translates the whole program	Translates one line at a time
Faster execution	Slower execution
Creates an executable file	No executable file
Reports all errors together	Stops at the first error
Example: GCC, G++	Example: Python, JavaScript

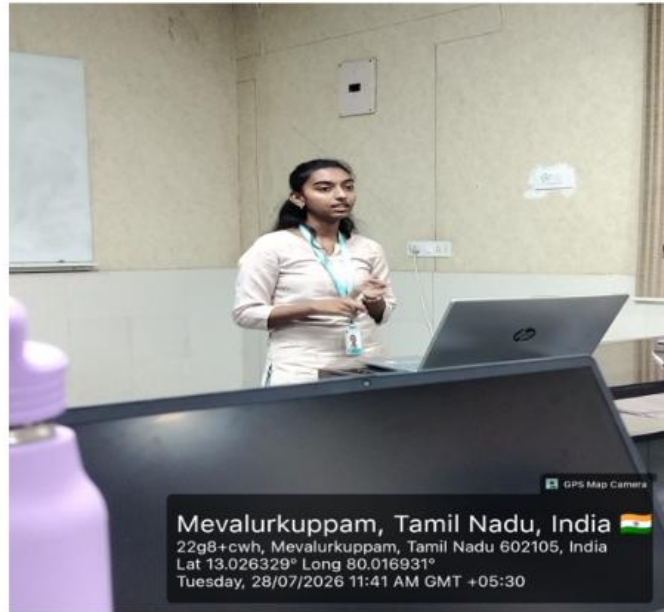


Conclusion



- Compiler is faster and suitable for large applications.
- Interpreter is easier to test and debug, making it ideal for scripting and rapid development.
- Both language processors convert high-level code into a form the computer can execute.

GEO TAG PHOTO



THANK YOU